

Requirement Engineering and Sustainability

Shola Oyedeji

Sustainability and Requirement Engineering (RE)



Environmental dimension: How do software systems, IT products, and services impact the environment?

Example: Energy usage, CO2 Emissions.

- Minimizing carbon footprint and optimizing energy usage.
- Reduce waste based on software usage



Individual dimension: How do software systems, IT products, and services support the individual endeavor toward that person's goals? **Example:** Support end - user adaptation, mental and physical well-being. Also, about supporting developer well-being.



Economic Dimension: The focus is on how to design and develop software systems, IT products, and services cost-effectively." Factors such as added value, profit and revenue are key.

- Balancing costs with long-term benefits and ROI

Social dimension: What are the impacts of software systems, IT products, and services on society? Example: the sense of belonging, interaction and social equity.

- Ensuring inclusivity, accessibility, and ethical considerations.

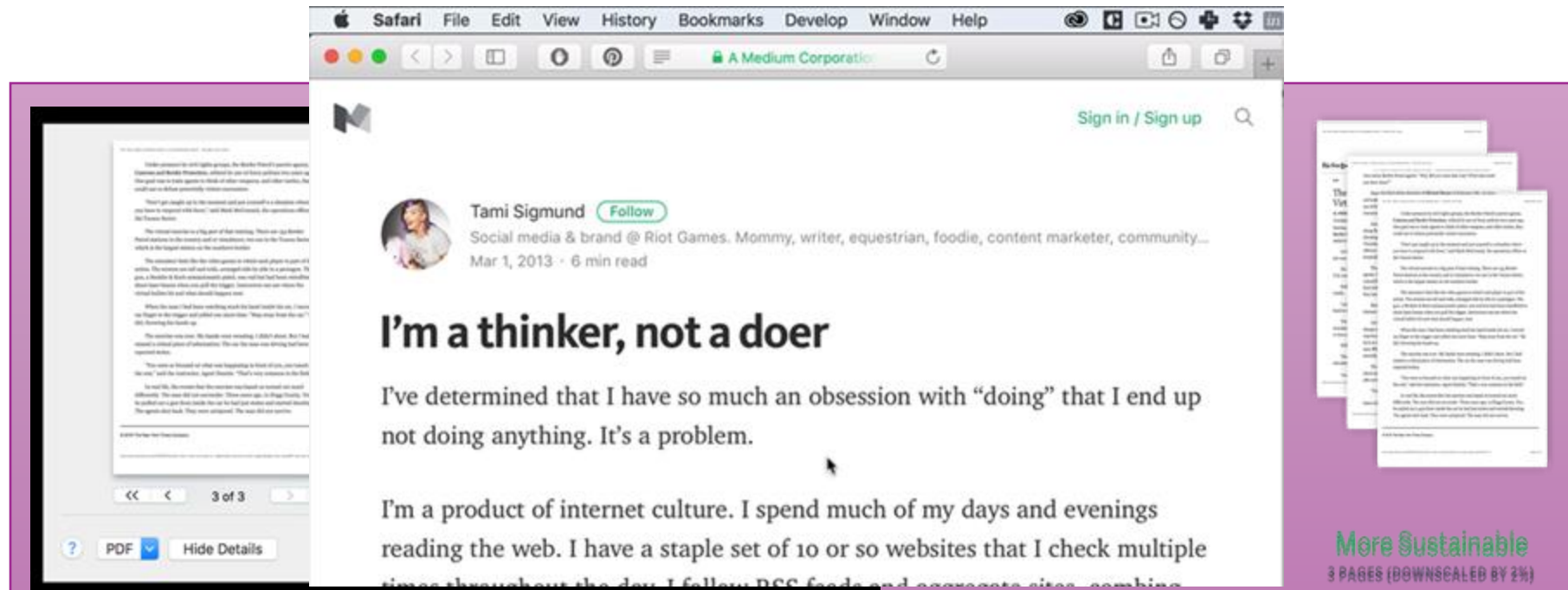


Technical Dimension: How can software systems, IT products, and services be designed and developed for easy evolution, maintainability, and adaptability to changes in the future?" Some factors: Designing maintainable, scalable, energy-efficient, usability systems.



Requirements and sustainability issues

- Mac OS on Paper waste
- Chrome don't allow scale down option but minimize margin options from more settings



Sustainability Awareness Framework (SusAF)



Sustainability Awareness Framework

A question-based “sustainability Awareness Framework” (SusAF) for raising awareness of the impacts that a software system could have on sustainability

SusAF aims to:

- Help identify the sustainability impacts and effects of IT products and services on the five dimensions of sustainability (social, individual, environmental, economic, and technical) across three orders of effect (direct, indirect, and systemic)
- Support discussion about the anticipated effects of IT products and services



SusAF Process



Key Elements in SusAF

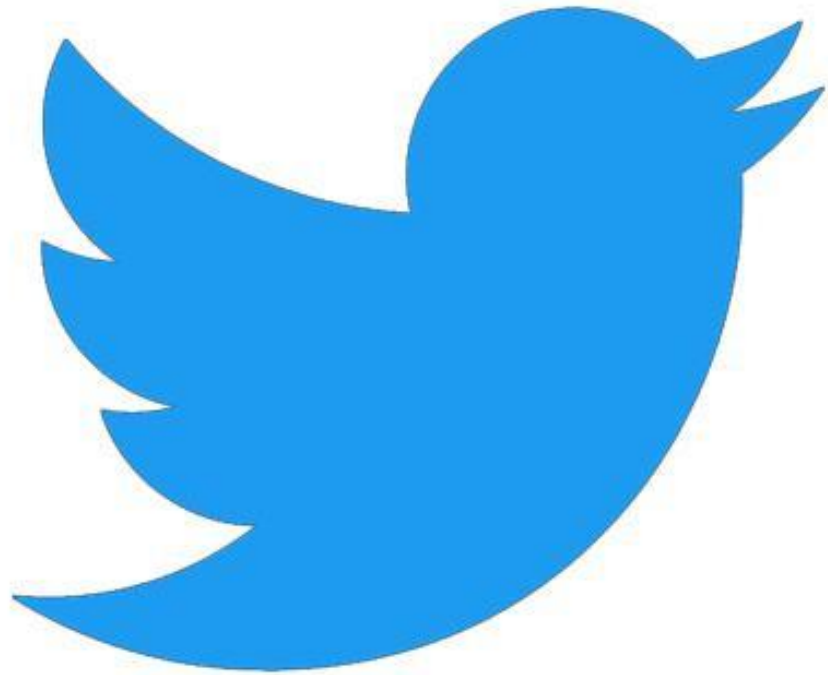
Social	(1) Sense of Community; (2) Trust; (3) Inclusiveness and Diversity; (4) Equality; (5) Participation and Communication;
Individual	(1) Health; (2) Lifelong learning; (3) Privacy; (4) Safety; (5) Self Awareness and Free will;
Environment	(1) Material and Resources; (2) Soil, Atmospheric and Water Pollution; (3) Energy; (4) Biodiversity and Land Use; (5) Logistics and Transportation;
Economic	(1) Value; (2) Customer Relationship Management (CRM); (3) Supply chain; (4) Ecosystem; (5)Governance and Processes; (6) Innovation and R&D;
Technical	(1) Maintainability; (2) Usability; (3) Extensibility and Adaptability; (4) Security; (5) Scalability;

Key Elements in SusAF

The assessment of sustainability effects on different time scale are classified into three order of impacts:

1. **First order (Immediate effects)** are the direct effects of the production, operation, use and disposal of socio-technical systems. This includes the properties and the full lifecycle impacts, such as in the Life-Cycle Assessment (LCA) approach. Although the timeline for the changes might be rather long (as we look at the lifecycle of the product), these effects are still direct consequences of the product.
2. **Second order (Enabling effects)** of operation and use of a system include any change enabled or induced by the system usage. These effects are not tied to the actual product but are side effects when the product is taken into use.
3. **Third order (Structural or systemic)** effects represent structural changes caused by the ongoing operation and use of the socio-technical system. These are usually consequences of changed habits due to the use of product and typically take a longer time to evolve. However, these effects are the game changers in sustainable development, effects that can truly make a difference.

Case Study



SusAF Scoping

Warm-Up



Introduce participants,
the SusAF,
and the IT product



Description of the IT **Product or Service**:

🕒 10 Min

The **X social media app** (formerly known as Twitter) offers a variety of functions designed to facilitate communication, information sharing, and interaction among its users.

List the **known sustainability effects** (SDG? CSR?):

🕒 10 Min

Key Elements in SusAF

Discuss & Select

Instructions

🕒 7 Min discuss & select for every single question

Discuss all the ideas within the team

- Decide which effects are worth capturing
- Paste in likelihood & impact matrix



Prioritize: Classify the effects using their likelihood and their level of impact



Dimensions: Social

Sense of community means the feeling of belonging to an organization, to an area or to a group of like-minded people.

- How can the product or service affect a person's sense of belonging to these groups?

Trust means having a firm belief in the reliability, truth, or ability of someone or something.

- How can the product or service change the trust between the users and the business that owns the system?

Inclusiveness and diversity refers to the inclusion of people who might otherwise be excluded or marginalized.

- How can the product or service impact on how people perceive others?
- What effects can it have on users with different backgrounds, age groups, education levels, or other differences?

Equity means the quality of being fair and impartial.

- How can the system make people to be treated differently from each other? (think data analytics or decision support)

Participation and communication refers to imparting or interchanging thoughts, opinions or information by speech, writing, or signs.

- How can the product or service change the way people:
 - › create networks?
 - › participate in group work?
 - › support, criticize or argue with others?

Key Elements in SusAF

Social Dimension

- Positive Impacts
 - Enable Information Sharing
 - Support social awareness
 - More inclusive society
- Negative Impacts
 - Influence consumer choices

Dimensions: Social

Sense of community means the feeling of belonging to an organization, to an area or to a group of like-minded people.

- How can the product or service affect a person's sense of belonging to these groups?

Trust means having a firm belief in the reliability, truth, or ability of someone or something.

- How can the product or service change the trust between the users and the business that owns the system?

Inclusiveness and diversity refers to the inclusion of people who might otherwise be excluded or marginalized.

- How can the product or service impact on how people perceive others?
- What effects can it have on users with different backgrounds, age groups, education levels, or other differences?

Equity means the quality of being fair and impartial.

- How can the system make people to be treated differently from each other?
(think data analytics or decision support)

Participation and communication refers to imparting or interchanging thoughts, opinions or information by speech, writing, or signs.

- How can the product or service change the way people:
 - › create networks?
 - › participate in group work?
 - › support, criticize or argue with others?

Key Elements in SusAF

Individual Dimension

- Positive Impacts
 - Enable communication
 - Aid User Knowledge
 - Enable education
- Negative Impacts
 - Cognitive overload
 - Witch Hunts
 - Mental distress

Dimensions: Individual

Health means the state of a person's mental or physical condition.

- How can the product or service improve or worsen a person's physical, mental, and/or emotional health?
- (For example, can it make a person feel anything good or bad - e.g. (under)valued, (dis)respected, (in)dependent, or coerced?)

Lifelong learning means the use of learning opportunities throughout people's lives for continuous development.

- How can the product or service affect people's competencies?

Privacy means being free from intrusion or disturbance in one's private life.

- How can the product or service expose (or help to hide) a person's identity, whereabouts or relations?

Safety means being protected from danger, risk, or injury.

- How can the product or service expose (or protect) a person from physical harm?
How can it make a person feel more (or less) exposed to harm?
- What if used in an unintended way?

Self-awareness and Free will means the capacity of an individual to act or make decisions on their own.

- How can the product or service empower (or prevent) a person from taking an action / decision when necessary?
- Can those affected by the product or service understand its implications, express concerns or be represented by someone?

Key Elements in SusAF

Environmental Dimension

- Positive Impacts
 - None
- Negative Impacts
 - Energy usage increases with more users
 - Depletion of resources
 - Giant data farms

Dimensions: Environmental

Material and resources includes everything that is needed to produce, deploy, operate, and cease a product or service.

How are materials consumed to produce the product or service?

What about to operate the product or service? E.g., requires hardware.

➤ How can it change the way people consume material? E.g., encourage to buy more?

Waste & pollution means effects the product or service might have on soil, atmospheric, and water pollution.

How can producing parts or supplies generate waste or emissions?

➤ How can the use itself produce waste or emissions?

How can it influence how much waste or emissions are generated?

➤ How can it promote (or impair) recycling?

Biodiversity includes the effects of a product or service on biodiversity in its operational environment and other affected land.

How can it impact the plants or animals around it? Or elsewhere?

➤ How can it change composition of the soil around it? E.g., occupying / cropland?

What about elsewhere?

Energy means all energy use that results from producing and using a product or service.

How can the product of service affect the need for production of energy?

What about the use of energy? E.g. encourages less energy.

➤ Does the hardware run on renewable energy? Is there a way to incentivise that?

Key Elements in SusAF

Economic Dimension

- Positive Impacts
 - Revenue generation
 - Easy Marketing
- Negative Impacts
 - Overspending from users

Dimensions: Technical

Maintainability means the degree to which an application is understood, repaired, or enhanced.

- How are the operating system and runtime environment expected to change what does that required from maintainers of this system?
- How can the correctness of the system be affected by other systems or affect the correctness of others?

Usability means the ability of users to productively use the system for the intended purpose.

- What kind of knowledge or physical properties are required to use the system and how can this affect different types of users? For example, is good eyesight and small, sensitive hands required to operate a system on a small handheld device?

Adaptability means the ability of a system to adapt itself to fit its behaviour according to changes in its environment or in parts of the system itself.

How could someone want to use the system in another context?
What can make that easier/more difficult?

- What can make that easier/more difficult for the system to adapt itself to fit new usage scenarios?

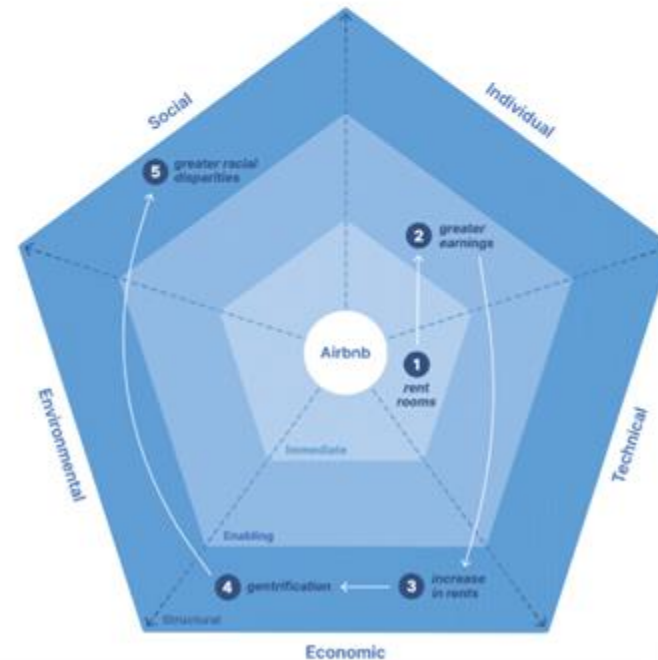
Security means freedom from, or resilience against, potential harm (or other unwanted coercive change) caused by external or internal attacks.

- Which assets controlled by this system would be desirable to an attacker?
E.g. financial information, people's whereabouts or preferences, etc.
What are the risks associated with these assets?
- What are other likely vulnerabilities of the system?

Key Elements in SusAF

Sustainability Awareness Diagram

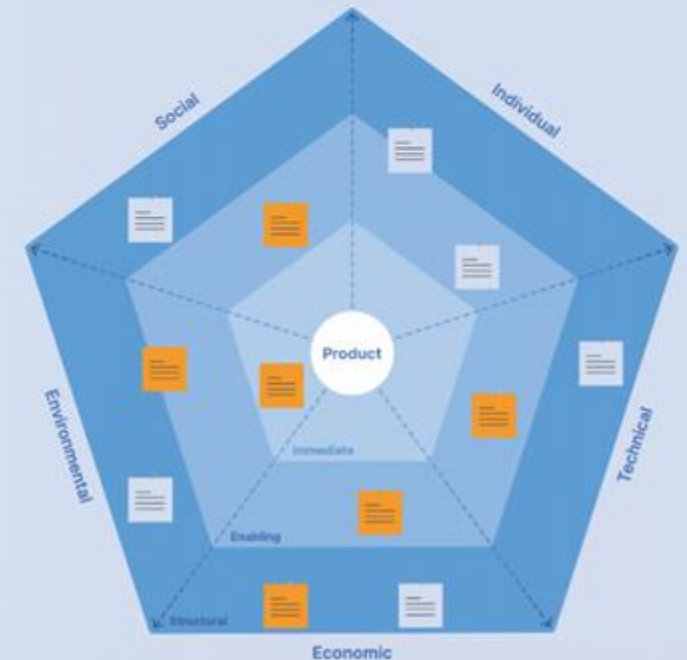
- › Is based on a radar chart
- › It is about cause and effects
 - › How do we get to a specific effect?
 - › What does this effect lead to?
- › Effects are placed in the dimensions and order of effects. They are connected with arrows (Cause and Effect); see Airbnb example.



Filling the SusAD

15 Min

1. Paste the effects from the high impact and likelihood corner of the matrix onto the SusAD according to dimension and order of effect.
2. Look at the remaining, less likely or less impactful, ones and choose which ones to still add to the SusAD (so it does not get too crowded).
3. Imagine your IT product or service is being used by many people over an extended period of time. What consequences may this have? And how do they relate?



- **SusAD:** a graphical illustration to visualize these impacts
- Write down effects/impacts of the IT product or service or software system features according to dimension and order of effect.

Key Elements in SusAF

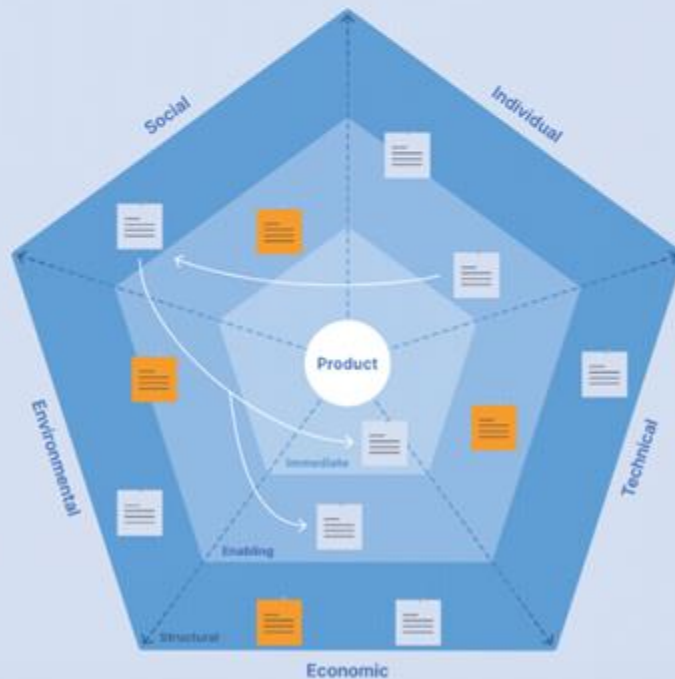
Chains of effects

Identify chains of effect:

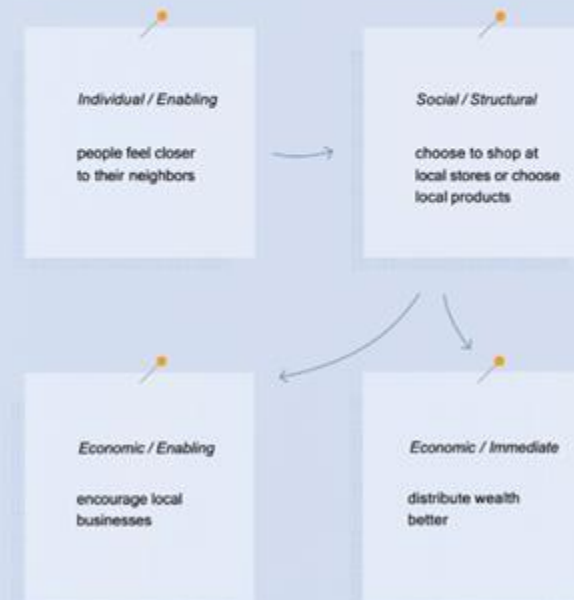
🕒 15 Min

Draw relations between the effects that may happen when many people use this product or service for several years:

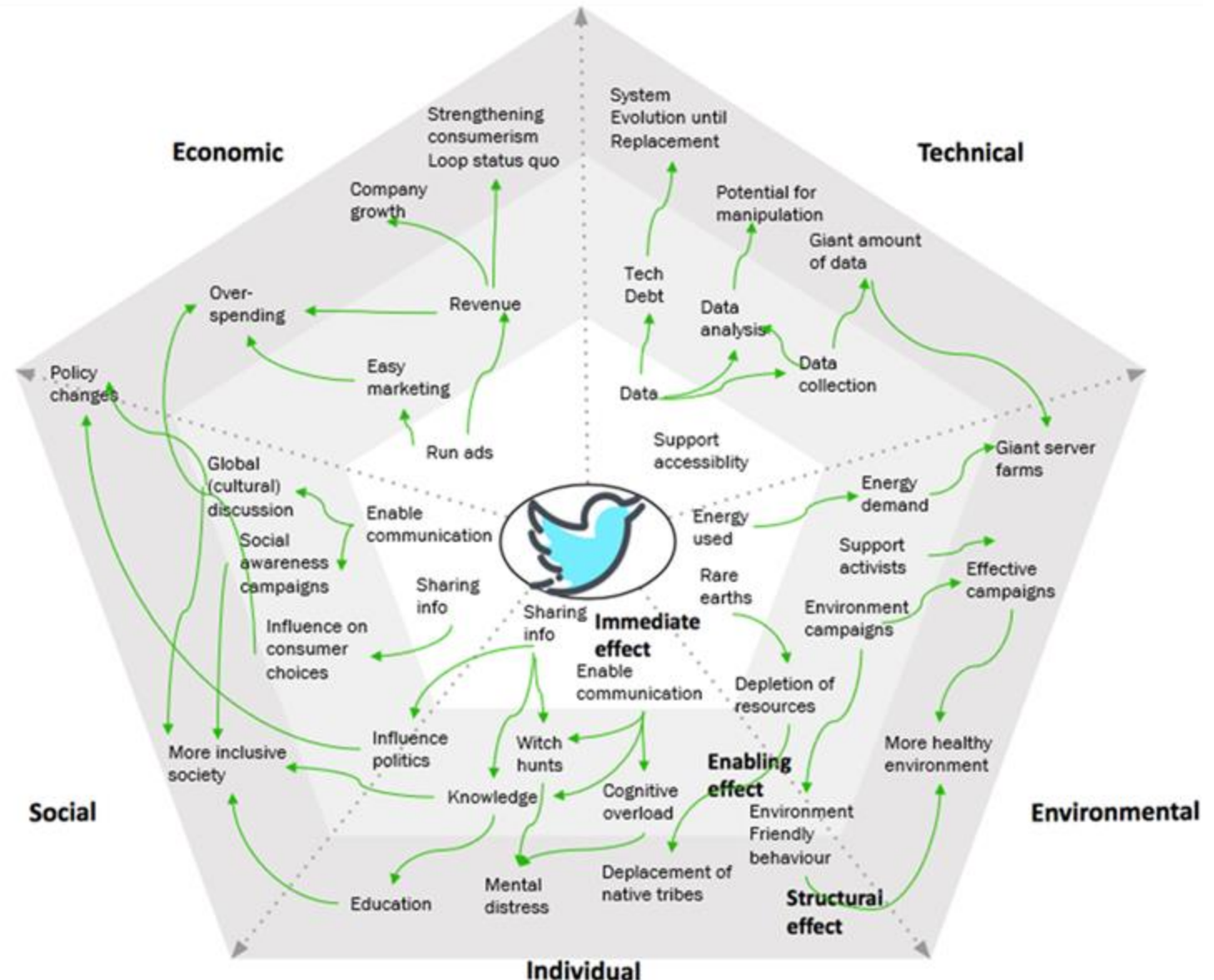
1. Think about which second order effects stem from which first order effects, and
2. Which third order effects can be a consequence of some second order effects
3. Effects can also have a related effect of the same order, and/or of a different dimension



Example: If people feel closer to their neighbors, they can choose to shop at local stores or choose local products, which can encourage local businesses and ultimately distribute wealth better



Key Elements in SusAF - Example



Analysis: SusAD

- Draw a relationship between effects that may happen when people use the IT product or services or software system

Key Elements in SusAF

Synthesis: Threats, opportunities, actions

- A structure to communicate sustainability impacts to diverse stakeholders.



Synthesis: Threats, opportunities, actions

Threats (Actions to Improve Negative Impacts):

- **Cognitive Overload:** Risk of users feeling overwhelmed by excessive information
 - Action: Implement personalized content filters and usage reminders.
- **Witch Hunts:** Risk of online harassment
 - Action: Enforce stricter community guidelines and AI-driven moderation.
- **Mental Distress:** Increased stress or anxiety from social media usage
 - Action: Introduce digital well-being tools and in-app support services.

Synthesis: Threats, opportunities, actions

Opportunities (Actions):

- Leverage platforms to create mental health resources and educational content
 - Action: Develop partnerships with mental health organizations to provide accessible tools and resources.
- Enhance user education through structured and verified knowledge-sharing features
 - Action: Introduce features like fact-checked educational threads and courses.

Writing Requirements with Sustainability Focus

Template for Functional Requirements:

- "The system shall [action] [sustainability impact]."
- **Example:** "The system shall adjust video resolution based on bandwidth to reduce energy usage."

Template for Non-Functional Requirements:

- "The system shall achieve [performance metric] to ensure [sustainability goal]."
- **Example:** "The system shall ensure a minimum of 70% renewable energy use in data centers to reduce environmental impact."

Non-functional requirements (NFRs)

Individual

1.User Personalization:

- The system shall provide personalized content filters and usage reminders to mitigate cognitive overload and enhance the user experience.

2.Content Validation:

- The platform shall ensure educational threads and courses are fact-checked and verified by recognized authorities to enhance knowledge sharing.

Social

1.Inclusivity and Diversity:

- The system shall support multilingual content translation for at least 90% of global languages to enable diverse and inclusive user communities.

2.Transparency in Advertising:

- The platform shall provide clear, transparent information on ads' origin, purpose, and targeting to avoid unethical influence on consumer choices.

Non-functional requirements

Technical

1.Accessibility:

- The platform shall comply with WCAG 2.1 standards, offering voice command functionality and screen reader support for all user interfaces.

2.Data Integrity:

- The system shall implement advanced data validation algorithms to detect and prevent the manipulation of user-generated content.

3.Technical Debt Management:

- The platform shall implement a bi-annual system refactoring process to address accumulated technical debt.

Environmental

1.Energy Efficiency:

- The platform's algorithms shall be optimized to reduce energy consumption by at least 15% annually, measured against baseline performance.

2.Sustainable Data Centers:

- All new data centers shall operate with renewable energy sources and implement energy-efficient cooling systems.

3.Resource Recycling:

- The system infrastructure shall incorporate 30% recycled hardware components to minimize the depletion of resources.

NFRs to Agile Requirement Gathering

User Story

Example of user story:

- Structure
- Acceptance Criteria

Task:

- Write a user story for a user who wants to search to find matching hotel option in a hotel web page.

User Story



As a <role>
I want <goal>
so that <benefit>

Acceptance criteria:
(Conditions of Satisfaction)

...

...

As an Account Manager
I want a sales report of my account
to be sent to my inbox daily
So that I can monitor the sales
progress of my customer portfolio

Acceptance criteria:

1. The report is sent daily to my inbox
2. The report contains the following sales details: ...
3. The report is in csv format.

Acceptance criteria (Gherkin format)

THE GIVEN/WHEN/THEN ACCEPTANCE CRITERIA: EXAMPLE 2

User story: As a user, I want to be able to request the cash from my account at an ATM so that I will be able to receive the money from my account quickly and in different places.

Scenario 1

Requesting the cash from a creditworthy account

Given

The account is creditworthy

And

The card is valid

And

The dispenser contains cash

When

The customer requests the cash

Then

Ensure the account is debited

And

Ensure cash is dispensed

And

Ensure the card is returned



altexsoft
software r&d engineering

THE GIVEN/WHEN/THEN ACCEPTANCE CRITERIA: EXAMPLE 2

User story: As a user, I want to be able to request the cash from my account at an ATM so that I will be able to receive the money from my account quickly and in different places.

Scenario 2

Requesting the cash from an overdrawn account

Given

The account is overdrawn

And

The card is valid

When

The customer requests the cash

Then

Ensure the rejection message is displayed

And

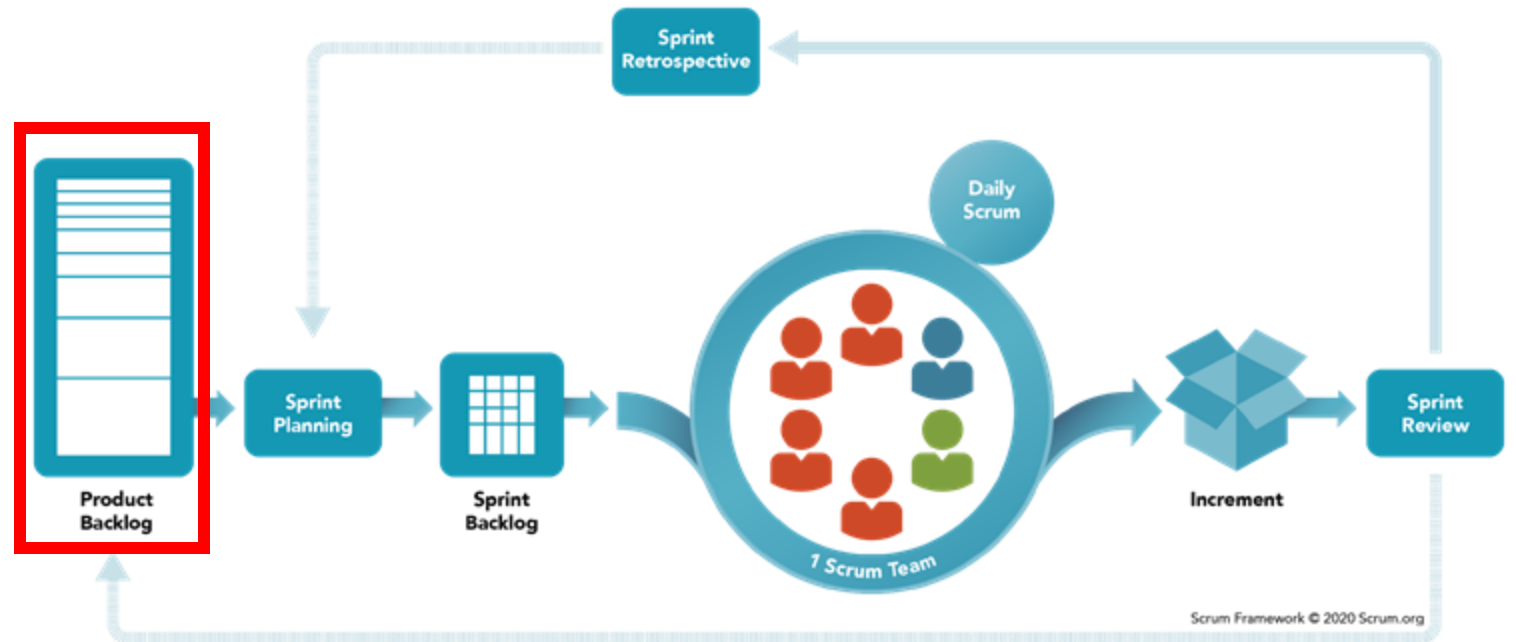
Ensure cash isn't dispensed



altexsoft
software r&d engineering

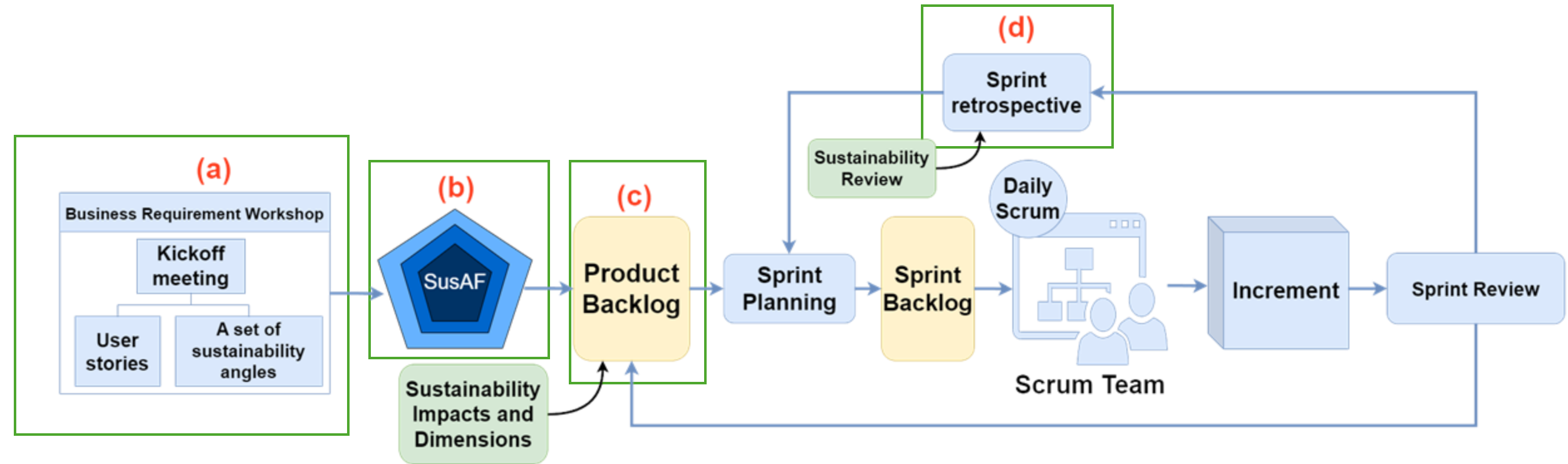
Scrum

- Scrum is one of the most famous agile methodologies characterized by cycles or stages of development called sprints.
- Scrum is a framework used by people to address complex adaptive problems



Industry Example

Planning → Action → Observation → Reflection



NFRs - Scrum Product Backlog

Converted Sustainability NFRs to Scrum Product Backlog Items

- For each NFRs
- Convert to user story
- Acceptance criteria (Gherkin format)
- Scrum product backlog Item (PBI) description

1. Identify the Sustainability Goal

Ask:

“What sustainability problem is this requirement solving?”

Example:

“Reduce cognitive overload” - Individual Sustainability

2. Translate the NFR into a User-Centered Goal

Every backlog item must be from a user’s perspective.

Convert:

The system shall...

to

As a user, I want... so that...

3. Measurable Acceptance Criteria (Gherkin)

Acceptance criteria must be:

- Testable
- Observable
- Measurable
- Binary (pass/fail)

Use:

Given / When / Then

Example:

Given usage time exceeds threshold - Then user receives reminder.

4. Define the PBI Scope

This includes:

- Description of the feature
- What is included and not included
- Sustainability value delivered
- Link to sprint or release goals

This keeps the Scrum team focused.

NFRs - Scrum Product Backlog

Individual dimension

1. User Personalization:

- The system shall provide personalized content filters and usage reminders to mitigate cognitive overload and enhance the user experience.

User Story

As a user, I want personalized content filters and usage reminders so that I can control my content exposure and avoid cognitive overload.

Acceptance Criteria

Given I am logged into my account

When I open the content settings

Then I should be able to enable/disable personalized content filters

Given I am actively using the platform

When my usage exceeds a configured threshold

Then I should receive a usage reminder notification

PBI Description

Develop personalized content filtering options and usage time reminders to support user well-being and reduce cognitive overload.

2.Content Validation:

- The platform shall ensure educational threads and courses are fact-checked and verified by recognized authorities to enhance knowledge sharing.

User Story

As a learner, I want educational threads to be verified by trusted authorities so that I can rely on accurate information.

Acceptance Criteria

Given a user publishes an educational thread

When it is submitted for publishing

Then the system checks it against approved fact-checking APIs or databases

Given validation fails

When the user views the thread status

Then the platform displays a “verification pending/failed” label

PBI Description

Implement automated and manual content validation workflows for educational posts, integrating verification mechanisms from recognized authorities.

NFRs → Scrum Product Backlog

Social Dimension

1. Inclusivity and Diversity:

- The system shall support multilingual content translation for at least 90% of global languages to enable diverse and inclusive user communities.

User Story

As a global user, I want the platform to translate content across many languages so that I can interact with diverse communities.

Acceptance Criteria

Given a post written in any supported language

When a user selects “Translate”

Then the platform should provide a translated version using the supported translation engine

Given translation fails or language is unsupported

When a user requests translation

Then a message should indicate the content cannot be translated

PBI Description

Extend language translation engine to support at least 90% of global languages and provide real-time translation for posts and comments.

2. Transparency in Advertising:

- The platform shall provide clear, transparent information on ads' origin, purpose, and targeting to avoid unethical influence on consumer choices.

User Story

As an end user, I want ads to display their origin, purpose, and targeting criteria so I can understand how my attention is being used.

Acceptance Criteria

Given an ad is displayed

When a user selects “Why am I seeing this?”

Then the system shows advertiser identity, targeting criteria, and ad purpose

Given targeting data is unavailable

When a user checks ad transparency

Then a clear message indicates the missing information

PBI Description

Implement an ad transparency module showing ad origin, purpose, and targeting to promote ethical advertising practices.

User Story

A good user story must follow the **INVEST** principle:

- **Independent:** Each user story should stand on its own, independent from other stories.
- **Negotiable:** Stories are not contracts, rather opportunities for negotiation and change.
- **Valuable:** Every story should add value for users and stakeholders.
- **Estimable:** Every story's time and budget costs should be calculable, based on domain and technical knowledge.
- **Small:** User stories should be small enough to estimate and implement simply.
- **Testable:** Make sure you can test the user story through criteria the story itself explains.



NFRs → Scrum Product Backlog

Task: Convert these into user stories for scrum product backlog:

Technical

1. Accessibility:

- The platform shall comply with WCAG 2.1 standards, offering voice command functionality and screen reader support for all user interfaces.

2. Data Integrity:

- The system shall implement advanced data validation algorithms to detect and prevent the manipulation of user-generated content.

3. Technical Debt Management:

- The platform shall implement a bi-annual system refactoring process to address accumulated technical debt.

Environmental

1. Energy Efficiency:

The platform's algorithms shall be optimized to reduce energy consumption by at least 15% annually, measured against baseline performance.

2. Sustainable Data Centers:

All new data centers shall operate with renewable energy sources and implement energy-efficient cooling systems.

3. Resource Recycling:

The system infrastructure shall incorporate 30% recycled hardware components to minimize the depletion of resources.

NFRs → Scrum Product Backlog

Technical

1. Accessibility:

- The platform shall comply with WCAG 2.1 standards, offering voice command functionality and screen reader support for all user interfaces.

Break into four smaller NFRs:

1. WCAG 2.1 compliance baseline
2. Keyboard navigation
3. Screen reader support
4. Voice command support

WCAG 2.1 Compliance Baseline

Refined NFR: The platform shall meet WCAG 2.1 Level AA guidelines for all core user flows (login, registration, posting content, reading content, and account settings).

User Story

As a user with accessibility needs, I want the platform to meet WCAG 2.1 Level AA so that I can use core features without barriers.

Acceptance Criteria (examples)

1. Color Contrast:

- Given I am viewing any text in the main interface
- When I use the default theme
- Then all text and interactive elements shall meet minimum contrast ratios defined in WCAG 2.1 Level AA.

2. Resizable Text

- Given I use browser zoom up to 200%
- When I navigate core pages (home, profile, settings, post view)
- Then content shall remain readable and functional without requiring horizontal scrolling.

NFRs → Scrum Product Backlog

Keyboard Navigation

Refined NFR

- All interactive elements (links, buttons, menus, inputs) shall be accessible and operable using only a keyboard.

User Story

- As a user who cannot use a mouse, I want to navigate the platform using only my keyboard so that I can access all features independently.

Acceptance Criteria

AC1. Tab Navigation

- Given I am on any core page
- When I press the Tab key
- Then focus shall move sequentially to all interactive elements in a logical order.

AC2. Visible Focus Indicator

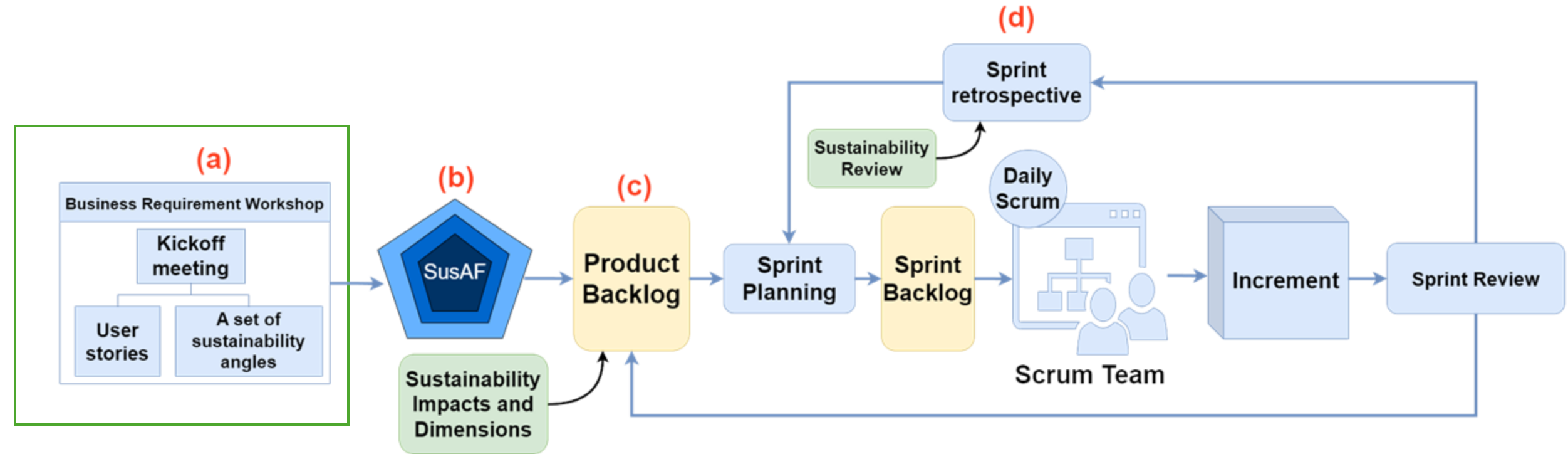
- Given I navigate using the keyboard
- When focus lands on an interactive element
- Then a visible focus indicator (e.g., outline) shall be shown.

AC3. Keyboard Activation

- Given I have focus on an interactive element
- When I press Enter or Space (as appropriate)
- Then the element shall activate (e.g., button click, link open, menu expand).

Industry Example

Planning → Action → Observation → Reflection



Sustainability Sprint Retrospective Template

What worked best?
What helped our sustainability design decision?

What were the obstacles and challenges?

What can we do differently to improve the overall sprint process?

What should we incorporate in the next sprint?

OBSERVING: What worked best and helped sustainability design decisions

OBSERVING: What were the obstacles and challenges

REFLECTING: What can we do differently to improve the overall sprint process?

PLANNING: What should be incorporated in the next sprint?

Sustainability Sprint Retrospective Example

Sprint	OBSERVING: What worked best and helped sustainability design decisions	OBSERVING: What were the obstacles and challenges	REFLECTING: What can we do differently to improve the overall sprint process?	PLANNING: What should be incorporated in the next sprint?
Sprint 1	One developer played the role of sustainability rep to keep the team on track as they made design decisions during this sprint.	It was challenging to use Jira and Excel templates at the same time. Team lacked sustainability knowledge and relied on SusAF, templates, and the researchers.	Incorporate the sustainability impacts into the sprint backlog. Establish some metrics to measure the sustainability of each sprint outcome.	Add sustainability impacts the sprint backlog. The researcher should join our sprint planning as a sustainability stakeholder.
Sprint 2	The continuous refactoring sessions helped reduce technical debt and keep the codebase clean. The cost-benefit analysis was useful to prioritize features that deliver maximum value	Despite efforts to optimize, the software product energy consumption is still higher than the desired consumption, especially during testing phases.	Identify and use better optimization strategies to improve the overall software product energy consumption.	Target reduction of energy consumption by 15% in the next sprint. Increase efforts to ensure diverse team participation in decision-making.
Sprint 3	The optimization of CI/CD pipelines reduced the energy consumption for the build and deployment processes.	Rapid iterations forced fast development and release, risking inefficient code results. This could cause technical debt and affect future maintainability	Create a better plan for each sprint iteration with a focus on supporting the team to create quality codes and test cases.	Work on efficient resource allocation to reduce delays in certain tasks that affect cost-effectiveness.
Sprint 4	The implementation of a modular architecture is good, which will aid in better long-term maintainability and adaptability in subsequent development iterations.	For individual sustainability, there is still a lack of personalization features, which might affect the ability of the software to meet the diverse needs of all users	Prioritize the implementation of personalization features for the software	Add at least two new personalization features for the next sprint. Add a green awareness campaign feature to educate users about sustainability. Add accessibility features for disabled users.

Questions?